SSN COLLEGE OF ENGINEERING

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

UEE2601- MICROPROCESSORS AND MICROCONTROLLERS -FUNDAMENTALS AND PRACTICES

Mini Project Report

Title : Temperature and Humidity Monitoring andControl System with Data Logging

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Year:2025-2026

			Marks Obtained			
S.NO	**Rubrics**	**Marks**				
1	Problem Specification	10				
2	Literature Survey	20				
3	Proposed Methodology ,Choice of hardware components with specification and Budget	30				
4	Hardware	10				
4	Presentation/Report/Viva (Team members contribution/ Work)	30				
	Total	100				

Remarks

Signature of the Faculty

1.Abstract

This project presents a Temperature and Humidity Monitoring System using the ESP32 microcontroller and DHT11 sensor. The system measures environmental parameters in real time and displays them on an LCD. It also logs the data at regular intervals onto an SD card for storage and future analysis. The proposed system is simple, cost-effective, and suitable for applications requiring continuous environmental monitoring..

2. Introduction

Monitoring environmental parameters such as temperature and humidity is essential in various fields including agriculture, healthcare, industrial processes, and smart systems. Accurate and continuous measurement of these parameters helps in maintaining optimal conditions, improving efficiency, and preventing damage to sensitive equipment or materials.

With the advancement of embedded systems, microcontrollers like the ESP32 have enabled the development of compact, low-cost, and efficient monitoring systems. The ESP32 offers high processing capability, built-in Wi-Fi and Bluetooth, and flexibility for interfacing with multiple sensors and modules.

In this project, a temperature and humidity monitoring system is developed using the DHT11 sensor and ESP32 microcontroller. The system provides real-time data display through an LCD and incorporates an SD card module for data logging at regular intervals. This allows users not only to monitor current conditions but also to analyze historical data.

The proposed system is simple, reliable, and can be further extended for automation and IoT-based remote monitoring applications.

3. Proposed Methodology

The proposed system is designed to monitor temperature and humidity in real time and store the data for future analysis. The system consists of an ESP32 microcontroller, DHT11 sensor, LCD display, and an SD card module.

Initially, the DHT11 sensor measures temperature and humidity from the surrounding environment. The sensed data is transmitted to the ESP32 microcontroller, which processes the values. The processed data is then displayed on the LCD screen for real-time monitoring.

Simultaneously, the ESP32 records the data at fixed intervals (e.g., every 2 seconds) and stores it in a CSV file on the SD card. Each data entry is logged along with a timestamp to maintain a proper record.

The system operates continuously, ensuring consistent data acquisition, display, and storage. This methodology provides a simple and efficient approach for environmental monitoring and can be further enhanced by integrating wireless communication for remote access and control.

4. Components(with Justification and Cost)

S.No	Component	Justification	Estimated Cost (₹)
1	ESP32 Microcontroller	Acts as the main controller to process sensor data, control display, and manage data logging. Supports SPI and I2C communication.	350 – 550
2	DHT11 Sensor	Measures temperature and humidity from the environment; easy to interface and low cost.	50 – 100
3	16×2 LCD with I2C Module	Displays real-time data; I2C reduces wiring complexity and uses fewer pins.	150 – 300
4	SD Card Module (SPI)	Enables data logging by storing sensor readings into files using SPI communication.	50 – 100
5	Micro SD Card	Provides storage for logged data and supports long-term monitoring.	300 – 400

Total Estimated Cost: ₹900 – ₹1450

5. Algorithm

- **Start the system**
- **Initialize components**
- Initialize ESP32 microcontroller
- Initialize DHT11 sensor
- Initialize LCD display

- Initialize SD card module
- **Check SD card**
- If SD card initialization fails, display error message
- Else, proceed to next step
- **Begin loop**
- **Read sensor data**
- Read temperature and humidity values from DHT11
- **Validate data**
- If reading is invalid, skip and read again
- **Display data**
- Show temperature and humidity on LCD
- **Get timestamp**
- Generate or fetch current time (for logging)
- **Store data**
- Write temperature, humidity, and timestamp to SD card in CSV format
- **Wait**
- Delay for 2 seconds
- **Repeat loop**
- Continue steps 5–10 continuously
- **End**
- continuously

6. Program

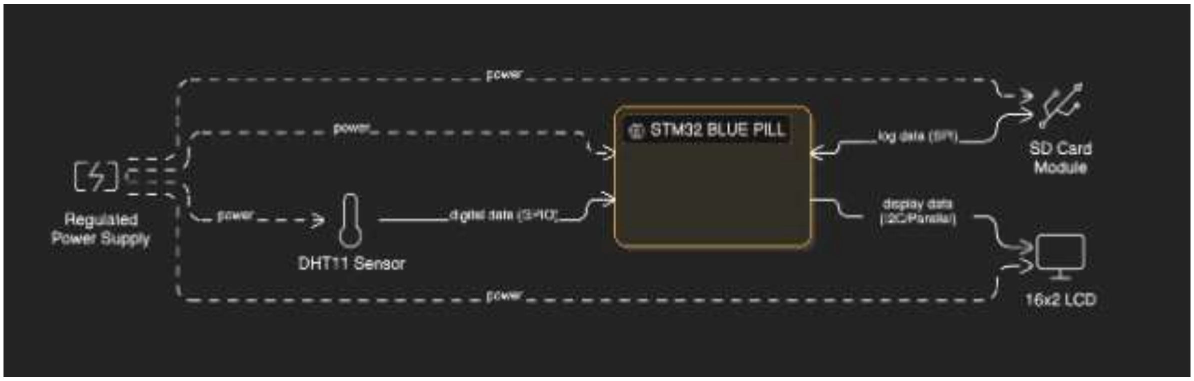
```
#include "DHT.h"
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SPI.h>
#include <SD.h>

#define DHTPIN 4
#define DHTTYPE DHT11
#define SD_CS 5

DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);

File dataFile;

void setup() {
  Serial.begin(115200);

  dht.begin();

  Wire.begin(21, 22);
```

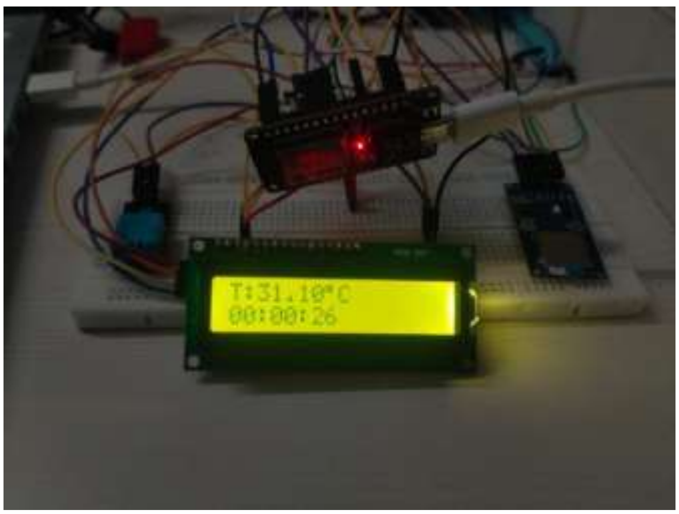
```
  lcd.init();
  lcd.backlight();

  lcd.print("Initializing...");

  if (!SD.begin(SD_CS)) {
    lcd.setCursor(0,1);
    lcd.print("SD Fail");
    return;
  }

  lcd.clear();
  lcd.print("SD Ready");
  delay(2000);
  lcd.clear();

  // File header
  dataFile = SD.open("/data.txt", FILE_APPEND);
  if (dataFile) {
    dataFile.println("Time,Temp(C),Humidity(%)");
    dataFile.close();
  }
}
```

```
void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();

  if (isnan(humidity) || isnan(temperature)) {
    lcd.setCursor(0,0);
    lcd.print("Sensor Error");
    delay(2000);
    return;
  }

  //Convert millis to HH:MM:SS
  unsigned long totalSeconds = millis() / 1000;
  int hours = totalSeconds / 3600;
  int minutes = (totalSeconds % 3600) / 60;
  int seconds = totalSeconds % 60;

  char timeString[9];
  sprintf(timeString, "%02d:%02d:%02d", hours, minutes, seconds);

  // LCD display
  lcd.setCursor(0,0);
```

```
  lcd.print("T:");
  lcd.print(temperature);
  lcd.print((char)223);
  lcd.print("C   ");

  lcd.setCursor(0,1);
  lcd.print(timeString);

  // Save to SD
  dataFile = SD.open("/data.txt", FILE_APPEND);
  if (dataFile) {
    dataFile.print(timeString);
    dataFile.print(",");
    dataFile.print(temperature);
    dataFile.print(",");
    dataFile.println(humidity);
    dataFile.close();
  }

  delay(2000); //
}
```

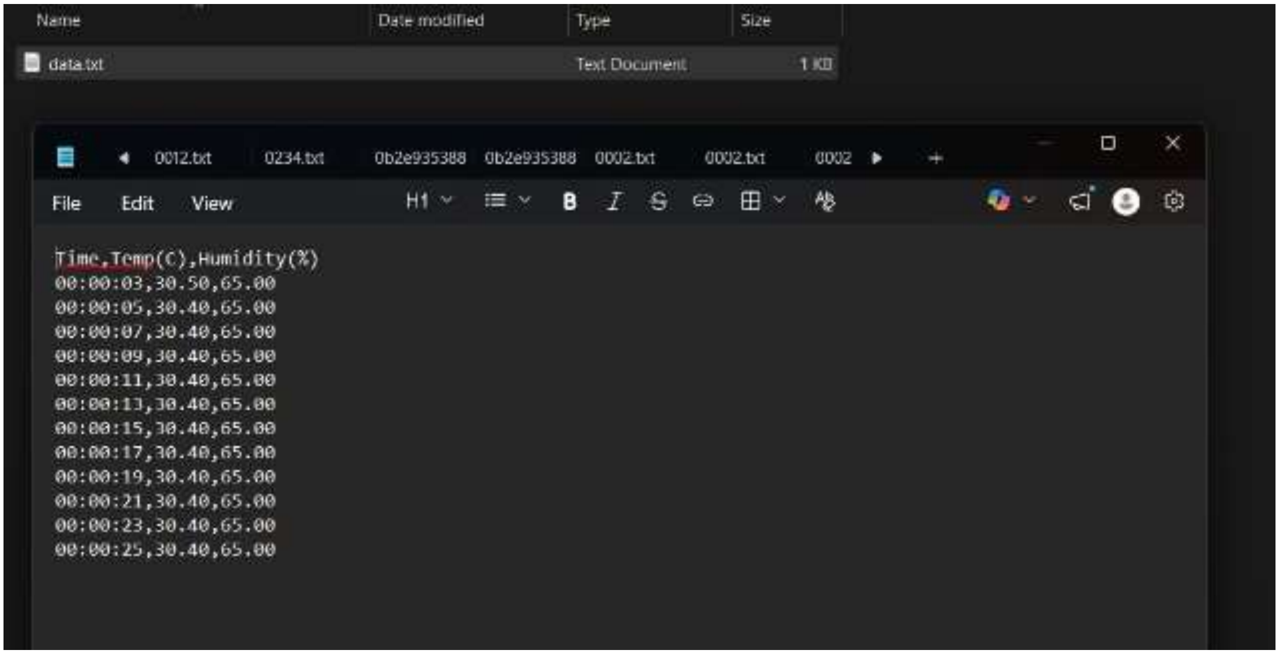
7. Architecture and Hardware Developed





The system is powered by a regulated power supply that provides stable voltage to allcomponents. The DHT11 sensor measures temperature and humidity and sends the data to theESP32 through a GPIO pin. The ESP32 processes this data and performs two main operations:displaying the real-time values on a 16x2 LCD and storing the data in an SD card using SPIcommunication. The system continuously monitors environmental conditions and logs the datafor future analysis.

Results are displayed in the LCD display and data logged in sd card.



8. Conclusion

The project successfully demonstrates the design and implementation of a temperature and humidity monitoring system using the ESP32 microcontroller. The system is capable of measuring environmental parameters in real time, displaying the data on an LCD, and storing it on an SD card for future analysis.

The integration of sensing, display, and data logging provides a complete and efficient monitoring solution. The use of ESP32 ensures reliable performance and allows for further expansion, such as IoT-based remote monitoring and automated control.

Overall, the system is cost-effective, simple to implement, and suitable for various applications like environmental monitoring, agriculture, and industrial use.

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9. References

1. Temperature and Humidity Monitoring System using ESP32 – Research Papers
2. ESP32 Microcontroller Datasheet
3. DHT11 Temperature and Humidity Sensor Datasheet
4. 16×2 LCD Display with I2C Module Documentation
5. SD Card Module (SPI Interface) Documentation
6. Arduino IDE Documentation
7. https://randomnerdtutorials.com/esp32-dht11-dht22-temperature-humidity-sensor-arduino-ide/
8. https://docs.espressif.com/projects/esp-idf/en/latest/esp32/
9. https://lastminuteengineers.com/esp32-dht11-dht22-web-server-tutorial/